

VPA ICT — IGCSE Y10S1 Weeks 10–18: Databases & Revision

Weeks 10 – 18: Databases & Revision

Syllabus: CIE IGCSE ICT (0417) Topic 15 + Paper 1 Revision | Software: Microsoft Access / LibreOffice Base

Week 10 — Relational Database Design

Syllabus: 15.1 Database Design

Keywords: relational database 关系型数据库 (guānxì xíng shùjùkù), entity 实体 (shí tǐ), attribute 属性 (shǔ xìng), primary key 主键 (zhǔ jiàn), foreign key 外键 (wài jiàn), relationship 关系 (guānxì), one-to-many 一对多 (yī duì duō), many-to-many 多对多 (duō duì duō), normalization 规范化 (guī fàn huà)

Activity 1 (15 min): "Why relational?" — Compare flat file (single table) vs. relational for a school system. What problems does repeating data cause?

Activity 2 (15 min): Entity identification — Given a school scenario, identify entities: Students, Teachers, Classes, Subjects. Draw entity relationship diagram.

Activity 3 (10 min): Primary key practice — For each entity, choose the best primary key. Why is "Name" a bad primary key?

Exam Tip: For database design questions, always identify: 1) Entities (tables), 2) Attributes (fields), 3) Primary keys (underlined), 4) Relationships (1:N or M:N). Missing any component loses marks.

Week 11 — Creating Tables & Relationships

Syllabus: 15.2 Creating Tables

Activity 1 (20 min): Create 3 tables in Access: Customers (CustomerID, Name, Email), Orders (OrderID, CustomerID, Date, Total), Products (ProductID, Name, Price).

Activity 2 (15 min): Set relationships — One Customer has Many Orders. Enforce referential integrity. Cascade updates.

Activity 3 (5 min): Test referential integrity — Try to delete a customer who has orders. What happens?

Week 12 — Queries: SELECT, WHERE, ORDER BY

Syllabus: 15.3 Queries

Keywords: query 查询 (chá xún), SELECT 选择 (xuǎn zé), WHERE 条件 (tiáo jiàn), ORDER BY 排序 (pái xù), AND 与 (yǔ), OR 或 (huò), BETWEEN 在...之间 (zài... zhī jiān), LIKE 模糊匹配 (mó hú pǐ pèi), wildcard 通配符 (tōng pèi fú)

Activity 1 (15 min): Simple queries — Show all customers from "Shenzhen." Show orders over \$100.

Activity 2 (15 min): Compound criteria — Orders over \$100 from Shenzhen. Names starting with "L" (LIKE "L*").

Activity 3 (10 min): Sorting — Sort customers by LastName ascending, then by FirstName descending.

SELECT CustomerName, Email FROM Customers WHERE City = "Shenzhen" ORDER BY CustomerName ASC;

Week 13 — Calculated Queries & Aggregates

Syllabus: 15.3 Advanced Queries

Keywords: calculated field 计算字段 (jìsuàn zìduàn), aggregate 聚合 (jùhé), SUM 求和 (qiúhé), AVG 平均 (píngjūn), COUNT 计数 (jìshù), GROUP BY 分组 (fēnzǔ), HAVING 分组条件 (fēnzǔ tiáojiàn)

Activity 1 (15 min): Calculated field — Create query showing OrderTotal with 10% tax: [Total]*1.10.

Activity 2 (15 min): Aggregate query — Total sales by month. Count orders by customer. Average order value.

Activity 3 (10 min): GROUP BY with HAVING — Show customers with more than 5 orders.

SELECT CustomerID, COUNT(OrderID) AS OrderCount FROM Orders GROUP BY CustomerID HAVING COUNT(OrderID) > 5;

Week 14 — Forms & Reports

Syllabus: 15.4 Forms and 15.5 Reports

Keywords: form 窗体 (chuāngtǐ), report 报表 (bàobiǎo), control 控件 (kòngjiàn), text box 文本框 (wénběn kuàng), label 标签 (biāoqiān), combo box 组合框 (zǔhé kuàng), subform 子窗体 (zǐ chuāngtǐ), header/footer 页眉/页脚 (yèméi/yèjiǎo)

Activity 1 (15 min): Create a data entry form for Customers with text boxes, labels, and navigation buttons.

Activity 2 (15 min): Create a sales report with: report header (title), page header (column labels), detail (data), group footer (subtotals), report footer (grand total).

Activity 3 (10 min): Subform — Customer form with subform showing their orders.

Week 15 — Database Integration with Web

Syllabus: 15.5 Integration

Activity 1 (15 min): Export Access table as HTML. View in browser. Discuss limitations.

Activity 2 (15 min): Export query results to Excel. Create chart from exported data.

Activity 3 (10 min): Paper 3 integration task practice — Import data from CSV → create database → query → export to HTML → display on web page.

Week 16 — Paper 1 Revision: Systems & Security

Activity 1 (20 min): Topic 1–4 rapid review — Hardware, software, storage, networks. Flashcard drill.

Activity 2 (15 min): Topic 9 security revision — Threats and countermeasures matching. Case study: a company loses data. What went wrong?

Activity 3 (5 min): Quick-fire questions: 10 questions, 30 seconds each.

Week 17 — Paper 1 Revision: Data, Effects & Systems Analysis

Activity 1 (20 min): Topic 5 data types — Binary, hexadecimal, ASCII. Convert A to binary (01000001). Convert 255 to hex (FF).

Activity 2 (15 min): Topic 6 effects — "Discuss the impact of e-commerce on traditional retail." Practice 6-mark answer structure.

Activity 3 (5 min): Topic 8 SDLC — Draw DFD for online booking system. Write pseudocode for login validation.

Week 18 — End of Semester Assessment

Mock Paper 3 (2h 30m):

Data Analysis (25 min): Import data, clean it, create formulas, generate 2 charts, create pivot table

Web Authoring (45 min): Create 3-page website with external CSS, table, form, navigation, images

Databases (50 min): Create 2 related tables, set relationships, create 3 queries (simple, compound, aggregate), design form and report

Integration (30 min): Export data from database to spreadsheet, create chart, embed in web page

Revision focus: All Paper 1 topics through 50 mixed questions.

Common Paper 3 Errors to Avoid

- ☐ Not setting primary keys before creating relationships
- ☐ Creating relationships without referential integrity
- ☐ Using wrong field type (e.g., number for phone number)
- ☐ Missing WHERE clause in update/delete queries
- ☐ Not testing web pages in browser after making changes
- ☐ Using spaces in filenames or HTML file paths
- ☐ Not saving evidence screenshots to designated folder
- ☐ Forgetting to print required evidence